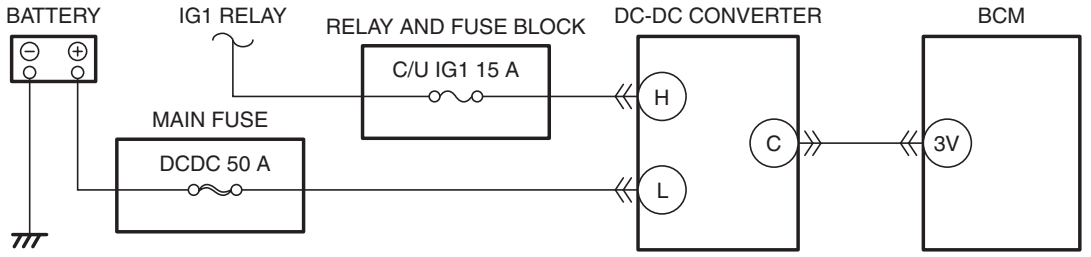
DTC U0298:68 [BCM]

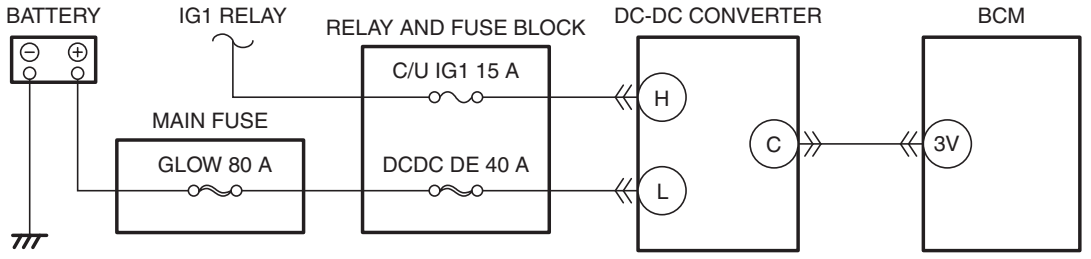
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Description	Communication error with DC-DC converter
Detection condition	• BCM could not receive the signals from the DC-DC converter for **5 s or more** with the ignition switched ON (engine off or on).
Fail-safe	Not applicable
Possible cause	• DC-DC converter connector terminal malfunction • DC-DC converter power supply circuit malfunction — Short to ground in wiring harness between the following: • DCDC 50 A fuse—DC-DC converter terminal L (SKYACTIV-G 1.3, SKYACTIV-G 1.5) • GLOW 80 A fuse—DC-DC converter terminal L (SKYACTIV-D 1.5) — DCDC 50 A fuse malfunction (SKYACTIV-G 1.3, SKYACTIV-G 1.5) — DCDC DE 40 A fuse malfunction (SKYACTIV-D 1.5) — GLOW 80 A fuse malfunction (SKYACTIV-D 1.5) — Open circuit in wiring harness between battery positive terminal and DC-DC converter terminal L • DC-DC converter power supply circuit malfunction — Short to ground in wiring harness between C/U IG1 15 A fuse and DC-DC converter terminal H — C/U IG1 15 A fuse malfunction — Open circuit in wiring harness between IG1 relay and DC-DC converter terminal H • BCM connector or terminal malfunction • Short to ground in wiring harness between DC-DC converter terminal C and BCM terminal 3V • Open circuit in wiring harness between DC-DC converter terminal C and BCM terminal 3V • DC-DC converter malfunction • BCM malfunction

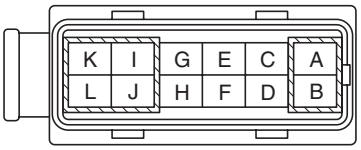
SKYACTIV-G 1.3, SKYACTIV-G 1.5

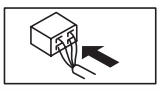


SKYACTIV-D 1.5

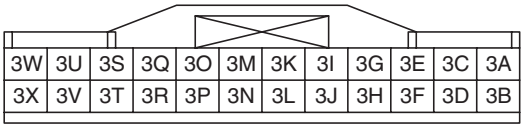


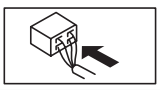
DC-DC CONVERTER
WIRING HARNESS-SIDE CONNECTOR





BCM
WIRING HARNESS-SIDE CONNECTOR





Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY BCM DTCs AGAIN** • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Go to the next step.
		No	Go to Step 10.
2	**INSPECT DC-DC CONVERTER CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the DC-DC converter connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 9.
3	**INSPECT DC-DC CONVERTER POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT OR SHORT TO GROUND** • Verify that the DC-DC connector is disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the DC-DC converter terminal L (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the DCDC 50 A fuse (SKYACTIV-G 1.3, SKYACTIV-G 1.5), DCDC DE 40 A fuse (SKYACTIV-D 1.5) and GLOW 80 A fuse (SKYACTIV-D 1.5). • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between the following: • DCDC 50 A fuse—DC-DC converter terminal L (SKYACTIV-G 1.3, SKYACTIV-G 1.5) • GLOW 80 A fuse—DC-DC converter terminal L (SKYACTIV-D 1.5) **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and DC-DC converter terminal L. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.

Step	Inspection		Action
4	**INSPECT DC-DC CONVERTER POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT OR SHORT TO GROUND** • Verify that the DC-DC connector is disconnected. • Switch the ignition ON (engine off or on). • Measure the voltage at the DC-DC converter terminal H (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the C/U IG1 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between C/U IG1 15 A fuse and DC-DC converter terminal H. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay and DC-DC converter terminal H. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
5	**INSPECT BCM CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the BCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 9.
6	**INSPECT DC-DC CONVERTER LIN COMMUNICATION CIRCUIT FOR SHORT TO GROUND** • Verify that the DC-DC converter and BCM connectors are disconnected. • Inspect for continuity between DC-DC converter terminal C (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter terminal C and BCM terminal 3V. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.

Step	Inspection		Action
7	**INSPECT DC-DC CONVERTER LIN COMMUNICATION CIRCUIT FOR OPEN CIRCUIT** • Verify that the DC-DC converter and BCM connectors are disconnected. • Inspect for continuity between DC-DC converter terminal C (wiring harness-side) and BCM terminal 3V (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter terminal C and BCM terminal 3V. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
8	**INSPECT DC-DC CONVERTER** • Inspect the DC-DC converter. (See DC-DC CONVERTER INSPECTION [WITHOUT i-ELOOP].) • Is the DC-DC converter normal?	Yes	Go to the next step.
		No	Replace the DC-DC converter, then go to the next step. (See DC-DC CONVERTER REMOVAL/INSTALLATION [WITHOUT i-ELOOP].)
9	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
10	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].)
		No	DTC troubleshooting completed.